# Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs 

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

| | Budding yeast | Nematode | Fruit fly | Mouse |
|---|---|---|---|---|
| **C002**<br>**akt-1**<br>4 genes / 3 species<br>5 direct pairs | No gene in this candidate network | **W1 akt-1**<br>GenAge 10<br>**GenAge annotation**<br>**Anti-Longevity**<br>Lifespan: Increase<br>Refs: 15998808, 23935948 | **F1 Akt1**<br>GenAge 832<br>**GenAge annotation**<br>**Pro-Longevity**<br>Lifespan: Decrease<br>Refs: 20976250 | **M1 Akt1**<br>GenAge 1810<br>**GenAge annotation**<br>**Anti-Longevity**<br>Lifespan: Increase<br>Refs: 23935948<br><br>**M2 Akt2**<br>GenAge 2231<br>**GenAge annotation**<br>**Anti-Longevity**<br>Lifespan: Increase<br>Refs: 28681509 |
| Direct pairs: W1–F1 (m:1); W1–M1 (m:n); W1–M2 (m:n); F1–M1 (1:n); F1–M2 (1:n) | | | | |
| **C104**<br>**ATG5**<br>3 genes / 3 species<br>2 direct pairs | **Y1 ATG5**<br>GenAge 1305<br>**GenAge annotation**<br>**Necessary for fitness**<br>Lifespan: Decrease<br>Refs: 20421943 | **W1 atg-5**<br>GenAge 897<br>**GenAge annotation**<br>**Pro-Longevity**<br>Lifespan: Decrease<br>Refs: 19469880 | No gene in this candidate network | **M1 Atg5**<br>GenAge 1862<br>**GenAge annotation**<br>**Pro-Longevity**<br>Lifespan: Increase<br>Refs: 23939249 |
| Direct pairs: W1–M1 (1:1); M1–Y1 (1:1) | | | | |
| **C005**<br>**ATG8**<br>3 genes / 3 species<br>3 direct pairs | **Y1 ATG8**<br>GenAge 1306<br>**GenAge annotation**<br>**Necessary for fitness**<br>Lifespan: Decrease<br>Refs: 20421943 | **W1 lgg-1**<br>GenAge 955<br>**GenAge annotation**<br>**Pro-Longevity**<br>Lifespan: Decrease<br>Refs: 18219227, 19469880 | **F1 Atg8a**<br>GenAge 22<br>**GenAge annotation**<br>**Pro-Longevity**<br>Lifespan: Increase + Decrease<br>Refs: 17617737, 18059160 19550147 | No gene in this candidate network |
| Direct pairs: W1–F1 (1:1); W1–Y1 (1:1); F1–Y1 (n:1) | | | | |

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.